

Assignment - 2

①

Given data:

$$\left. \begin{array}{l} \text{Power} = 300 \text{ kW} \\ \text{Supply freq.} = 60 \text{ Hz} \end{array} \right\} \text{ requirement.}$$

Supply freq. available = 50 Hz.

Case ① without an electronic circuit / Automation.

To obtain 60 Hz from 50 Hz $\rightarrow$ use a motor-generator set to convert the frequency;

Reqn: of speed;

$$N_s = \frac{120f}{P}$$

$N_s = N_{s_m} = N_{s_g}$ (speed remains same for f_m & f_g)
(synchronous speed)

$$\Rightarrow N_{s_m} = \frac{120f_m}{P_m} ; N_{s_g} = \frac{120f_g}{P_g}$$

$$\Rightarrow \frac{120f_m}{P_m} = \frac{120f_g}{P_g}$$

$$\Rightarrow \frac{f_m}{f_g} = \frac{P_m}{P_g}$$

$$= \frac{50}{60} = \frac{P_m}{P_g} = \underline{\underline{\frac{5}{6}}}$$

$$\Rightarrow \frac{P_m}{P_g} = \frac{10}{12}$$

$$\text{ie } P_m : P_g = 10 : 12$$

means without automation / electronic circuit,

$f_m \rightarrow$ freq. of the motor
 $P_m \rightarrow$ no. of poles of motor
 $P_g \rightarrow$ no. of pole of gen.
 $f_g \rightarrow$ freq of generator.

$$f_m = 50 \text{ Hz}$$

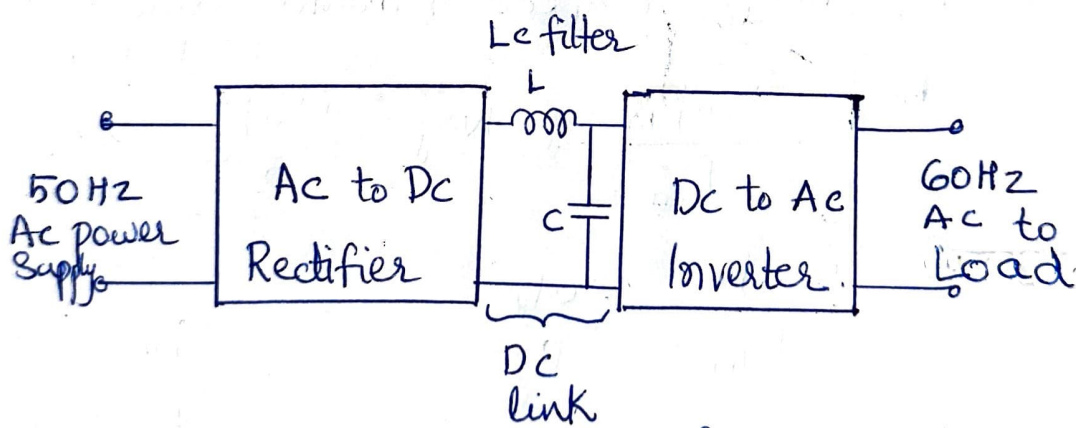
$$f_g = 60 \text{ Hz}$$

A motor-generator set with motor of 12 poles and 50Hz frequency and generator with 10 poles and 60Hz frequency can serve the purpose.

Case (2) With electronic circuit / automation:

By using a Ac-DC-Ac Converting System a 50Hz ac can be converted to a 60Hz ac power supply for the load.

A 3phase fully controlled rectifier is used to convert the 50Hz ac power into a dc power. Further, it is converted to 60Hz 3ph ac power by a Dc-Ac Converter by controlling the switching frequency of the switches to ^{output} 60Hz Ac power.



- * Ac to Dc Rectifier $\rightarrow$ 3ph ^{fully} Controlled Rectifier
- * Dc to Ac Inverter $\rightarrow$ 3ph fully controlled inverter
- * Dc link, Lc filter $\rightarrow$ Lc filter, filters out the current and Voltage ripples.

Qn: 2

Firing angle calculation:

$$V_{avg} = \frac{2 V_m \cos \alpha}{\pi}$$

$$V_L = 440 \text{ V}$$

$$V_{ph} = \frac{440}{\sqrt{3}} = 254 \text{ V.}$$

$$V_{ph} = \frac{V_m}{\sqrt{2}} = 359.2 \text{ V}$$

$$(V_m = V_{ph} \times \sqrt{2}).$$

$$\text{As } V_f = 100 \text{ V,}$$

$$V_{f avg} \Rightarrow 100 \text{ V} = \frac{2 \times 359.2}{\pi} \times \cos \alpha$$

$$\Rightarrow \alpha = \cos^{-1} \left(\frac{100 \times \pi}{2 \times 359.2} \right)$$

$$= \cos^{-1} (0.437)$$

$$\boxed{\alpha = 64.09^\circ}$$

firing angle $\alpha = 64.09^\circ$

③

Given data :

$$f_{NL} = 51 \text{ Hz}$$

$$\text{slope} = \frac{1 \text{ MW}}{1 \text{ Hz}} \quad (s_p)$$

$$\text{Load-1} = 1000 \text{ kW}$$

$$\text{Load-2} = 800 \text{ kW}$$

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Operating frequency of system before switch is closed :

* only Load-1 = 1000 kW is connected.
(1 MW)

$$\begin{aligned} * \text{ frequency drop from N.load} &= \frac{P}{s_p} = \frac{1 \text{ MW}}{(1 \text{ MW/Hz})} \\ &= \underline{\underline{1 \text{ Hz}}} \end{aligned}$$

$$\begin{aligned} * \text{ operating frequency} &= f_{NL} - f_{\text{drop}} \\ &= 51 - 1 \\ &= \underline{\underline{50 \text{ Hz}}} \end{aligned}$$

②

Operating frequency after Load-2 is connected.

$$\begin{aligned} * \text{ Total load} &= \text{Load 1} + \text{load 2} = 1800 \text{ kW} \\ &= \underline{\underline{1.8 \text{ MW}}} \end{aligned}$$

$$* \text{ frequency drop} = \frac{P}{s_p} = \frac{1.8 \text{ MW}}{1 \text{ MW/Hz}} = \underline{\underline{1.8 \text{ Hz}}}$$

$$* \text{ operating frequency} = 51 - 1.8 = \underline{\underline{49.2 \text{ Hz}}}$$

c) Action to restore the frequency to 50Hz?

* Operator must increase the governor set point to increase 49.2Hz to 50Hz by 0.8Hz frequency.

* As per the slope $S_p = 1 \text{ MW/Hz}$,

Change in freq. req. = 0.8Hz

change in P should be; $\Delta P = S_p \times 0.8 \text{ Hz}$

$$= 1 \text{ MW/Hz} \times 0.8 \text{ Hz}$$

$$= \underline{\underline{0.8 \text{ MW}}}$$

* Thus, the operator must increase the prime mover input power by 0.8MW by increasing the governor set point.